# Results

# Results I

This section reports repository facts produced by the meta-project's live introspection pass. Test counts, module inventories, pipeline-stage counts, and figure data are injected from generated artifacts. Runtime and achieved-coverage claims are intentionally omitted when no versioned benchmark or coverage artifact backs them.

## Multi-Project Pipeline Execution I

The `./run.sh` interactive orchestrator can execute the public roster through the shared DAG. The table below is a structural snapshot, not a timing benchmark: counts come from repository introspection, and the coverage column states each exemplar's declared policy floor rather than an unversioned observed percentage.

# Multi-Project Pipeline Execution I

| Project | Effective core stages[1] | Discovered tests | Declared project floor |
|---|---|---|---|
| `template_code_project` | 8 | 258 | 90% |
| `template_prose_project` | 8 | 165 | 90% |
| `template_autoresearch_project` | 8 | 318 | 90% |

## Multi-Project Pipeline Execution I

[1]"Core-only" excludes LLM-tagged and other opt-in stages according to the YAML stage tags. A fresh run must be used to establish completion status or wall-clock performance for a particular machine and dependency set.

# Infrastructure Test Suite I

| Metric | Value |
|---|---|
| Test files | 997+ |
| Total tests | ~13,093 |
| Infrastructure coverage gate | $\geq$60% configured floor |
| Prohibited mock-framework imports | Checked by the static no-mocks gate |

## Infrastructure Test Suite I

Exercises Pandoc/XeLaTeX paths, steganography hashing, telemetry, YAML-driven pipeline DAG resolution, HTTPS-bound optional suites (`pytest-httpserver`), and local Ollama-gated subsets.

## Infrastructure Module Inventory I

The introspection module (`template_template.core.introspection`) emits the authoritative table below—every row reflects `discover_infrastructure_modules(REPO_ROOT)`.

# Infrastructure Module Inventory I

| Module | Python Files | Has AGENTS.md | Has README.md | Key Exports |
|---|---|---|---|---|
| `autoresearch` | 11 | ✓ | ✓ | `build_autoresearch_plan`, readiness validation CLI |
| `benchmark` | 4 | ✓ | ✓ | Template harness scoring + comparative gates |
| `config` | 0 | ✓ | ✓ | Repository defaults + hardened templates |
| `core` | 202 | ✓ | ✓ | `get_logger`, `load_config`, `TemplateError` |
| `docker` | 0 | ✓ | ✓ | Containerisation scaffolding |
| `doctor` | 14 | ✓ | ✓ | Checkout diagnose/fix/undo repairs |
| `documentation` | 23 | ✓ | ✓ | `FigureManager`, `build_api_index`, `generate_markdown_table` |
| `fonds` | 6 | ✓ | ✓ | — |

# Infrastructure Module Inventory I

| Module | Python Files | Has AGENTS.md | Has README.md | Key Exports |
|---|---|---|---|---|
| `formal_contracts` | 5 | ✓ | ✓ | — |
| `llm` | 57 | ✓ | ✓ | Ollama helpers, sanitization, review + translation pipelines |
| `logrotate.d` | 0 | ✓ | ✓ | Rotation snippets (documentation-first) |
| `metadata` | 3 | ✓ | ✓ | `normalized_repository_url`, `generate_metadata_package` |
| `methods` | 10 | ✓ | ✓ | `build_methods_orchestration_plan`, methods-stage contracts + validation |
| `orchestration` | 14 | ✓ | ✓ | `PipelineRunner`, entry point for `./run.sh` |
| `project` | 123 | ✓ | ✓ | `discover_projects`, workspace management |
| `prose` | 9 | ✓ | ✓ | Markdown readability + prose tooling |

# Infrastructure Module Inventory I

| Module | Python Files | Has AGENTS.md | Has README.md | Key Exports |
|---|---|---|---|---|
| `provenance` | 9 | ✓ | ✓ | — |
| `publishing` | 89 | ✓ | ✓ | Zenodo, executable bundle, archival targets |
| `reference` | 16 | ✓ | ✓ | BibTeX models, parsers, converters |
| `rendering` | 123 | ✓ | ✓ | PDF/HTML/slide rendering, Pandoc filters |
| `reporting` | 416 | ✓ | ✓ | Coverage parsers, dashboards, executive artefacts |
| `research` | 3 | ✓ | ✓ | — |
| `rules` | 6 | ✓ | ✓ | — |
| `scientific` | 4 | ✓ | ✓ | `check_numerical_stability`, `benchmark_function` |

# Infrastructure Module Inventory I

| Module | Python Files | Has AGENTS.md | Has README.md | Key Exports |
|---|---|---|---|---|
| `search` | 77 | ✓ | ✓ | `infrastructure.search.literature` clients + cache |
| `sia` | 10 | ✓ | ✓ | Self-Improving-AI loop: task validation, harness, metric capture |
| `skills` | 8 | ✓ | ✓ | `discover_skills`, SKILL manifest regeneration |
| `steganography` | 14 | ✓ | ✓ | Watermark overlays + hash manifests |
| `tools` | 6 | ✓ | ✓ | — |
| `transmission` | 6 | ✓ | ✓ | — |
| `validation` | 87 | ✓ | ✓ | PDF + Markdown + integrity CLIs |

## Infrastructure Module Inventory I

All 31 enumerated subdirectories carry Tier-1/`README.md` and Tier-2/`AGENTS.md` coverage wherever the Documentation Duality standard applies; subsets ship Tier-3 `SKILL.md` descriptors for MCP routing (`infrastructure/skills` manifest generation).

# Agentic Skill Documentation Coverage I

| Layer | Role |
| --- | --- |
| Shared agent guidance | Root `AGENTS.md`, `.agents/` skills, roles, and workflows |
| Structural | `AGENTS.md` directories |
| Skills | Optional `SKILL.md` manifests + generated `.agents/skill_manifest.json` |
| PAI capsule | Repository level `PAI.md` narratives |

## Agentic Skill Documentation Coverage I

385+ Markdown shards under `docs/` capture operational knowledge without duplicating auto-generated inventories.

## DAG Reference (Declarative Executor) I

The table summarizes the 10 default full-run stages (core plus live LLM nodes) from 17 YAML declarations; core mode selects 8. Separately tagged opt-in extensions are omitted here. Entries follow dependency order rather than numeric script filenames.

# DAG Reference (Declarative Executor) I

| Name | Typical script / method | Responsibility | Failure semantics |
|---|---|---|---|
| Clean Output Directories | `_run_clean_outputs` | Deletes stale `output/` trees | Blocking |
| Environment Setup | `scripts/pipeline/stage_00_setup.py` | Validates tooling, PYTHONPATH scaffolding | Blocking |
| Infrastructure Tests | `scripts/pipeline/stage_01_test.py --infra-only --infra-scope pipeline-smoke` | Focused infrastructure smoke contracts | Required tests must pass |
| Project Tests | `scripts/pipeline/stage_01_test.py --project-only` | Project-local tests + coverage gates | Required tests and coverage must pass |
| Project Analysis | `scripts/pipeline/stage_02_analysis.py` | Executes `projects/<name>/scripts/*.py` | Blocking |
| PDF Rendering | `scripts/pipeline/stage_03_render.py` | Pandoc → XeLaTeX manuscripts | Blocking |
| Output Validation | `scripts/pipeline/stage_04_validate.py` | Structural PDF/markdown probes | Blocking / warnings |
| LLM Manuscript Review | `scripts/pipeline/stage_06_llm_review.py --reviews-only` | Local Ollama reviews | Fresh policy-bound skip; required live acceptance remains blocking |

# DAG Reference (Declarative Executor) I

| Name | Typical script / method | Responsibility | Failure semantics |
|---|---|---|---|
| LLM Translations | `scripts/pipeline/stage_06_llm_review.py --translations-only` | Optional translations | Fresh policy-bound skip |
| Copy Outputs | `scripts/pipeline/stage_05_copy.py` | Mirrors deliverables → `output/<project>/` | Blocking |

# DAG Reference (Declarative Executor) I

`scripts/pipeline/stage_07_executive_report.py` provides separate **multi-project executive reporting**, not an additional single-project DAG node (`execute_pipeline.py`). The multi-project executor writes its aggregate summary after project execution.

## Steganographic Performance Boundary I

The steganography path exposes metadata injection, SHA-256 hashing, overlay generation, and optional QR-code insertion as separately testable operations. This revision makes no latency claim because the project does not currently generate a versioned benchmark artifact recording hardware, input PDF, warm-up policy, repetitions, and dispersion. A future performance table should be compiled from such an artifact rather than copied from an ad-hoc workstation run.

Rendered via `projects/templates/template_template` (`generate_manuscript_metrics.py` → injected tokens such as 31). Architecture figures stem from `template_template.figures.architecture_viz`—font sizes constrained by §QA.

# Infrastructure · 1 / 8

| | |
|---|---|
| **autore search**<br>11 py<br>[ARS·] | **benchmark**<br>4 py<br>[ARS·] |
| **config**<br>0 py<br>[ARS·] | **core**<br>204 py<br>[ARS·] |

# Infrastructure · 2 / 8

| | |
|---|---|
| **docker**<br>0 py<br>[ARS·] | **doctor**<br>14 py<br>[ARS·] |
| **documen tation**<br>23 py<br>[ARS·] | **fonds**<br>6 py<br>[ARS·] |

# Infrastructure · 3 / 8

| | |
|---|---|
| **formal contracts**<br>5 py<br>[ARS·] | **llm**<br>57 py<br>[ARS·] |
| **logrotate.d**<br>0 py<br>[AR··] | **metadata**<br>3 py<br>[ARS·] |

# Infrastructure · 4 / 8

| | |
|---|---|
| **methods**<br>10 py<br>[ARS·] | **orchest ration**<br>14 py<br>[ARS·] |
| **project**<br>123 py<br>[ARS·] | **prose**<br>9 py<br>[ARS·] |

# Infrastructure · 5 / 8

| | |
|---|---|
| **provenance**<br>9 py<br>[ARS·] | **publishing**<br>89 py<br>[ARS·] |
| **reference**<br>16 py<br>[ARS·] | **rendering**<br>125 py<br>[ARS·] |

# Infrastructure · 6 / 8

| | |
|---|---|
| **reporting**<br>418 py<br>[ARS·] | **research**<br>3 py<br>[ARS·] |
| **rules**<br>6 py<br>[ARS·] | **scientific**<br>4 py<br>[ARS·] |

# Infrastructure · 7 / 8

| | |
|---|---|
| **search**<br>77 py<br>[ARS·] | **sia**<br>10 py<br>[ARS·] |
| **skills**<br>8 py<br>[ARS·] | **stegano graphy**<br>15 py<br>[ARS·] |

# Infrastructure · 8 / 8

**tools**

6 py
[ARS·]

**transm ission**

6 py
[ARS·]

**validation**

87 py
[ARS·]

# Projects · 1 / 7

| | |
|---|---|
| **active inference**<br>16 chapters<br>84 test files | **advanced literature review**<br>23 chapters<br>8 test files |
| **autopoiesis**<br>8 chapters<br>31 test files | **autore search project**<br>6 chapters<br>23 test files |

# Projects · 2 / 7

| | |
|---|---|
| **autosci entists**<br><br>9 chapters<br>14 test files | **code project**<br><br>9 chapters<br>16 test files |
| **data descriptor**<br><br>8 chapters<br>7 test files | **eda notebook**<br><br>9 chapters<br>12 test files |

# Projects · 3 / 7

| | |
|---|---|
| **experiment tree**<br>6 chapters<br>5 test files | **formal**<br>7 chapters<br>34 test files |
| **gold refinement**<br>12 chapters<br>25 test files | **literature meta analysis**<br>21 chapters<br>74 test files |

# Projects · 4 / 7

| | |
|---|---|
| **madlib**<br><br>12 chapters<br>8 test files | **methods paper**<br><br>9 chapters<br>12 test files |
| **newspaper**<br><br>6 chapters<br>10 test files | **pitch deck**<br><br>7 chapters<br>14 test files |

# Projects · 5 / 7

**pools rules tools**

8 chapters
13 test files

**prose project**

8 chapters
13 test files

**redacted report**

3 chapters
7 test files

**registered report**

8 chapters
5 test files

# Projects · 6 / 7

| | |
|---|---|
| **search project**<br><br>9 chapters<br>33 test files | **sia**<br><br>6 chapters<br>16 test files |
| **storybook**<br><br>6 chapters<br>7 test files | **template**<br><br>21 chapters<br>11 test files |

# Projects · 7 / 7

**textbook**

51 chapters
13 test files

Figure 3: Live rendering of the Two-Layer Architecture from repository introspection: the infrastructure layer (top) holds the 31 enumerated infrastructure directories, each annotated with its Python file count and a four-slot documentation badge—A AGENTS.md, R README.md, S SKILL.md, P PAI.md, with · marking an absent file—so a fully documented module reads [ARSP]. A YAML DAG arrow connects it to the project

## Self-Referential Analysis I

layer (bottom) of public exemplars labelled with chapter and test-file counts. The takeaway: documentation-duality coverage is near-uniform across the infrastructure, and every box was placed from the same live data the prose cites.

# Stages 1–5 of 17

| | | |
|---|---|---|
| **1** | Clean Output Directories | core clean |
| **2** | Environment Setup | core |
| **3** | Infrastructure Tests | core tests |
| **4** | Project Tests | core tests |
| **5** | Project Analysis | core |

# Stages 6–10 of 17

| | | |
|---|---|---|
| **6** | Connector Search | science live |
| **7** | Provenance Record | provenance |
| **8** | PDF Rendering | core |
| **9** | Output Validation | core |
| **10** | LLM Manuscript Review | llm live |

# Stages 11–15 of 17

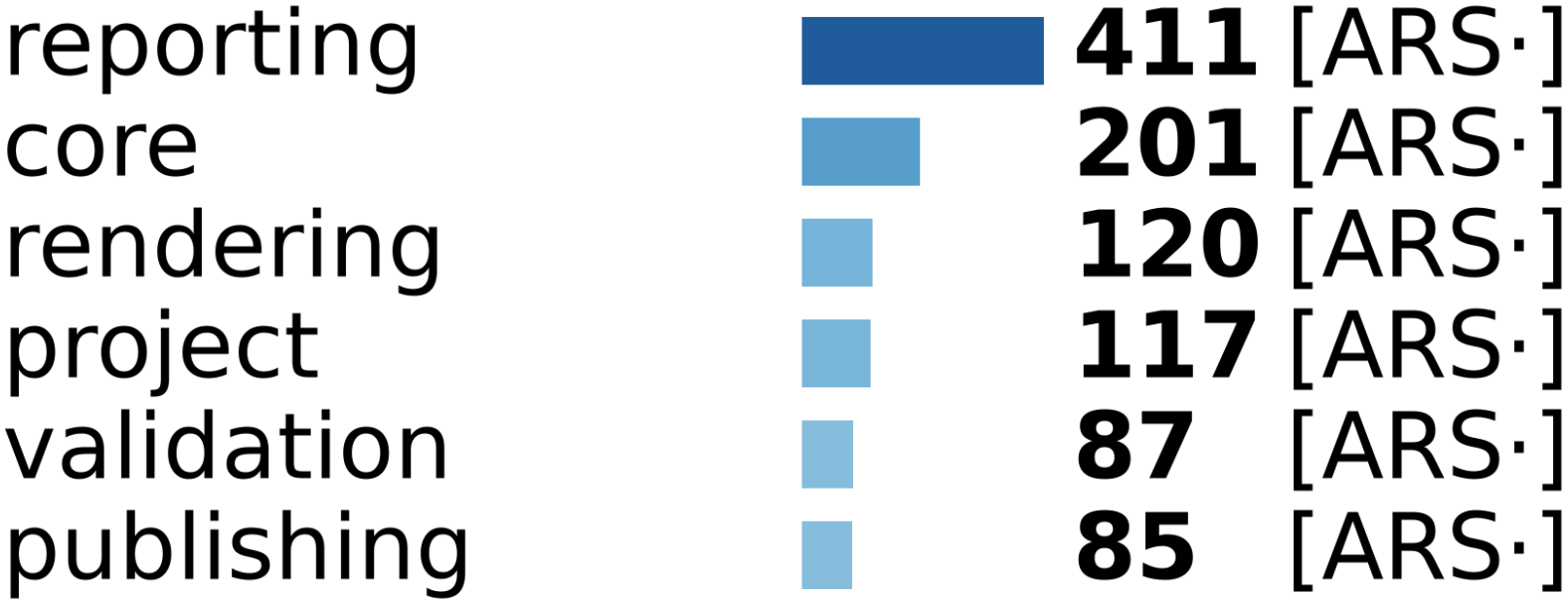
| | | |
|---|---|---|
| **11** | LLM Translations | llm<br>live |
| **12** | Copy Outputs | core |
| **13** | Ebook Generation | core<br>ebook |
| **14** | docxplus Export | core<br>docxplus |
| **15** | Metadata Package | core<br>metadata |

# Stages 16–17 of 17

| | | |
|---|---|---|
| **16** | Executable Bundle | bundle |
| **17** | Archival Publication | archival live |

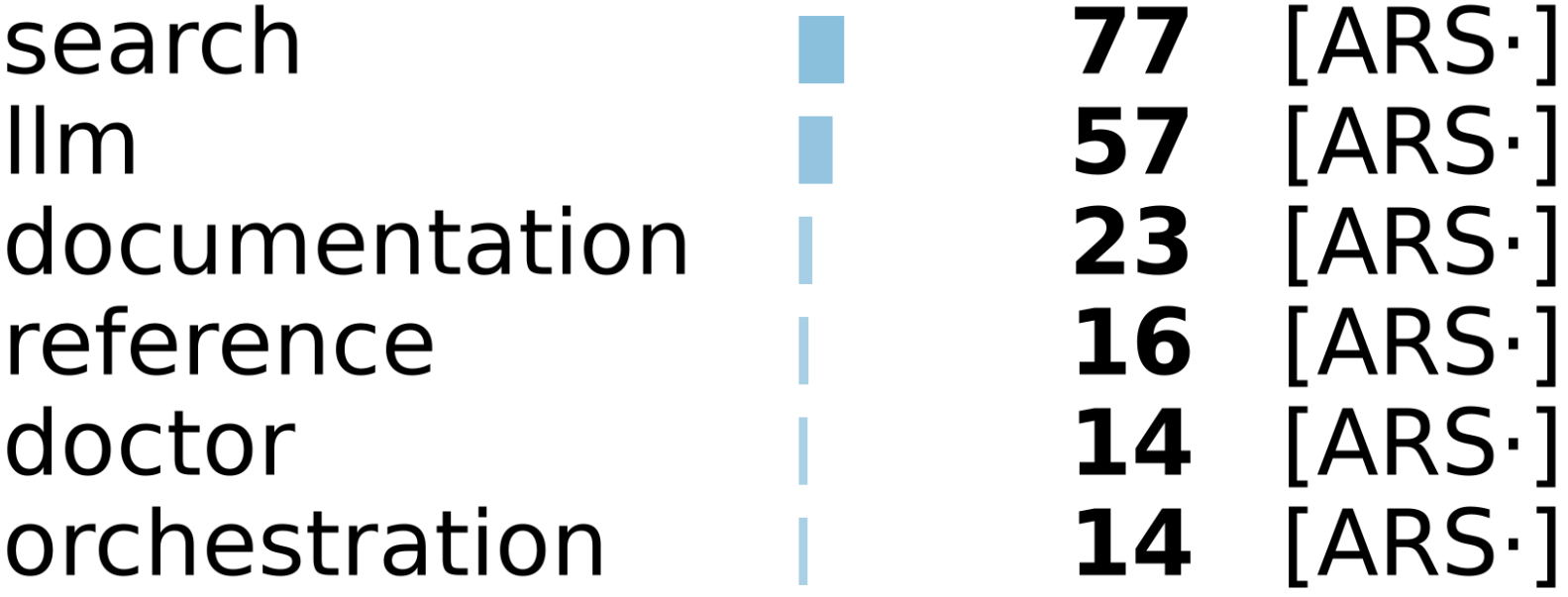
Figure 4: Declared-stage reference with all 17 YAML stages in reading order, showing complete names and tags (core, LLM, ebook, docxplus, metadata, bundle, archival). Panel order documents declarations; it does not imply a linear dependency chain.

## Modules 1–6 of 31




Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

## Modules 7–12 of 31




Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

## Modules 13–18 of 31

| | | |
|---|---|---|
| steganography | **13** | [ARS·] |
| autoresearch | **11** | [ARS·] |
| methods | **10** | [ARS·] |
| sia | **10** | [ARS·] |
| prose | **9** | [ARS·] |
| provenance | **9** | [ARS·] |

Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

## Modules 19–24 of 31

| | | |
|---|---|---|
| skills | **8** | [ARS·] |
| fonds | **6** | [ARS·] |
| rules | **6** | [ARS·] |
| tools | **6** | [ARS·] |
| transmission | **6** | [ARS·] |
| formal_contracts | **5** | [ARS·] |

Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

## Modules 25–30 of 31

| | | |
|---|---|---|
| benchmark | **4** | [ARS·] |
| scientific | **4** | [ARS·] |
| metadata | **3** | [ARS·] |
| research | **3** | [ARS·] |
| config | **0** | [ARS·] |
| docker | **0** | [ARS·] |

Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

**Modules 31–31 of 31**

logrotate.d **0** [AR··]

Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

Figure 5: Horizontal file-count histogram of every infrastructure subdirectory, sorted largest-first.

## Self-Referential Analysis I

The long tail of small, single-purpose packages beside a handful of larger ones (`core`, `validation`, `publishing`) is the visual signature of the Unix-philosophy modularity the architecture section argues for—capability concentrated where it compounds, not spread evenly by fiat.

# Comparative Feature Analysis I

The comparative feature matrix below summarizes the Appendix F capability matrix.

## Tool key · A–J

A template/

B Snakemake 9.x

C Nextflow 25.x

D CWL 1.2

E Quarto 1.x

F Jupyter Book 2.x

G R Markdown

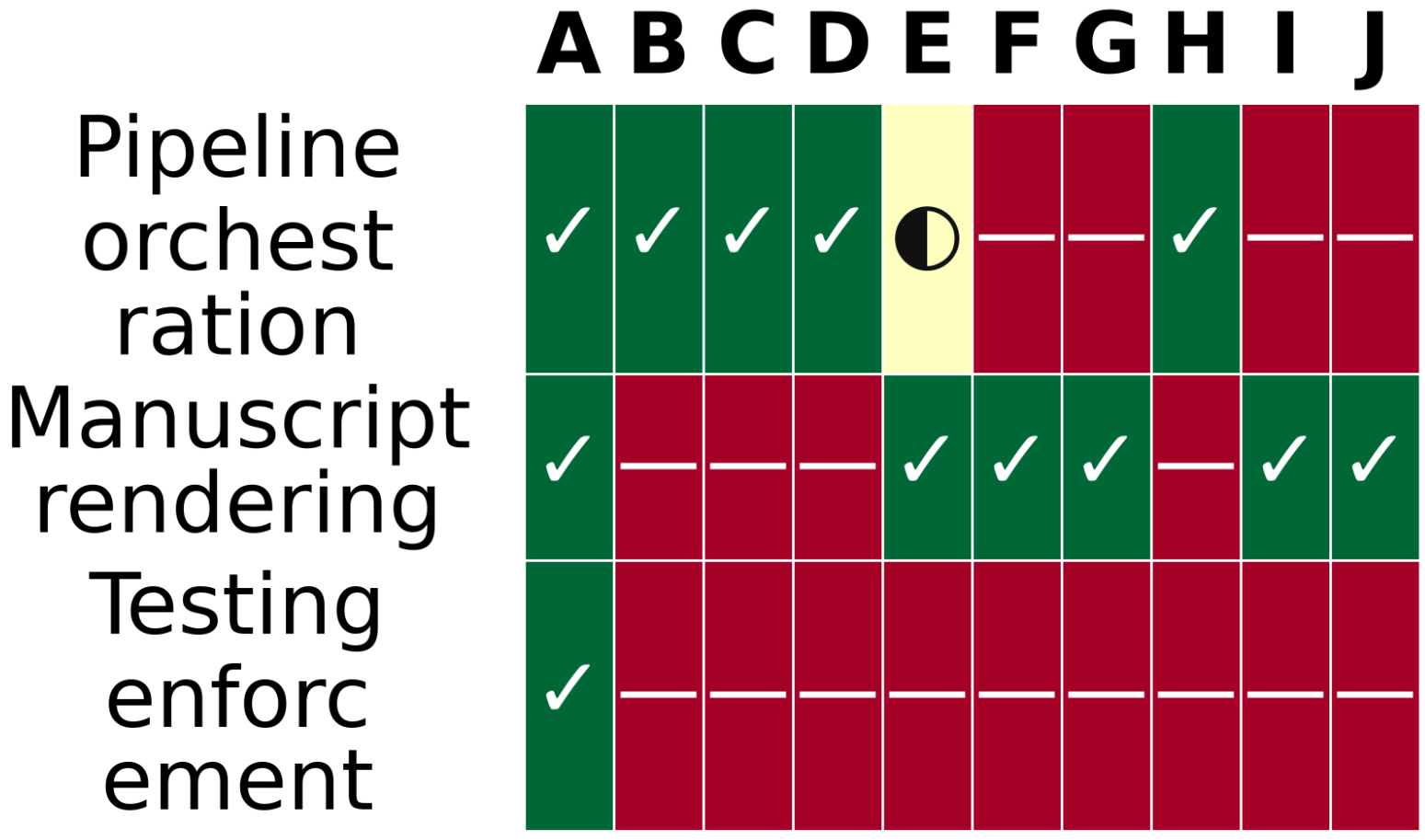
H DVC 3.x

I Overleaf (2025)

J OpenAI Prism

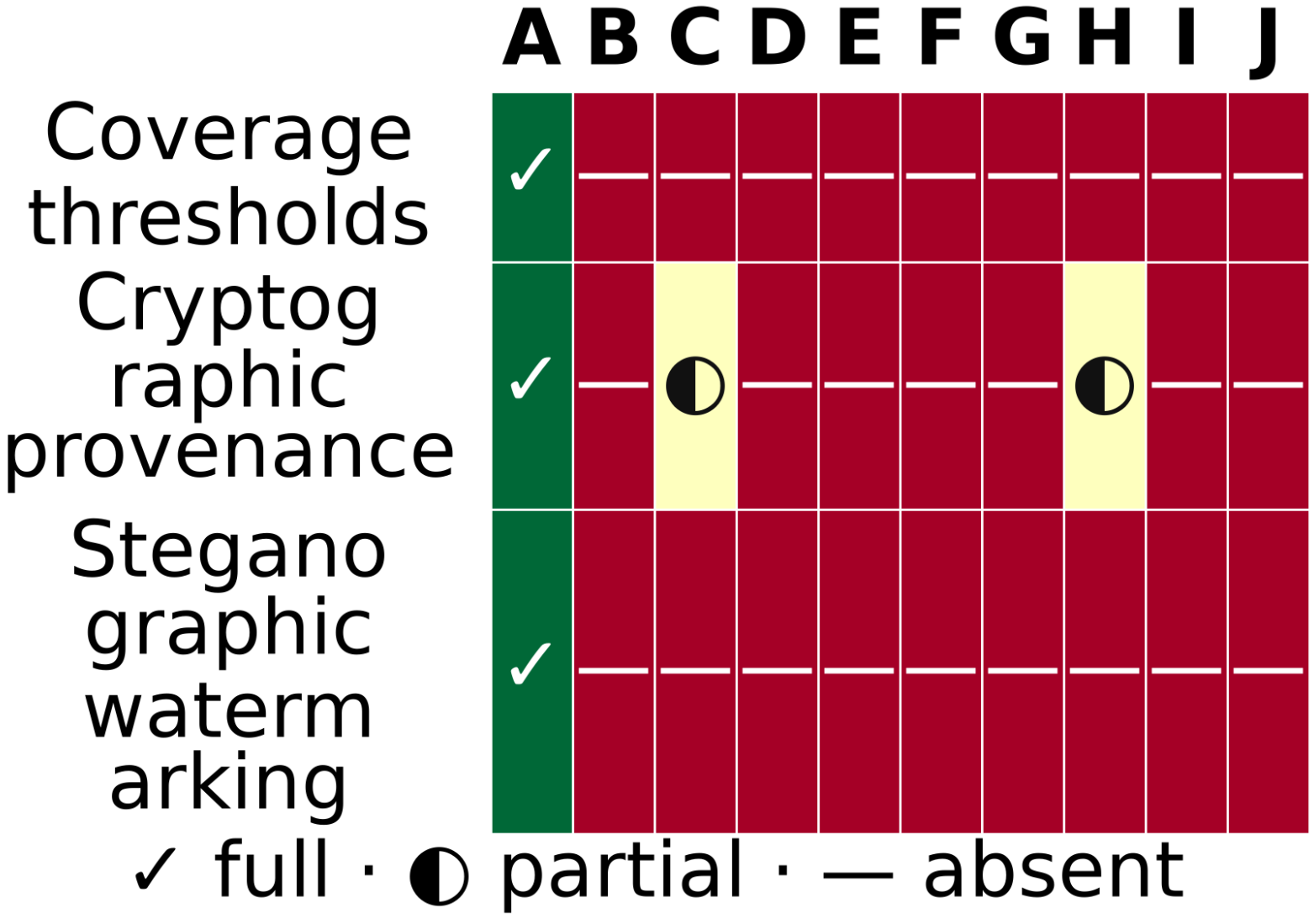
✓ full · ◐ partial · — absent

# Capabilities 1–3 · tools A–J

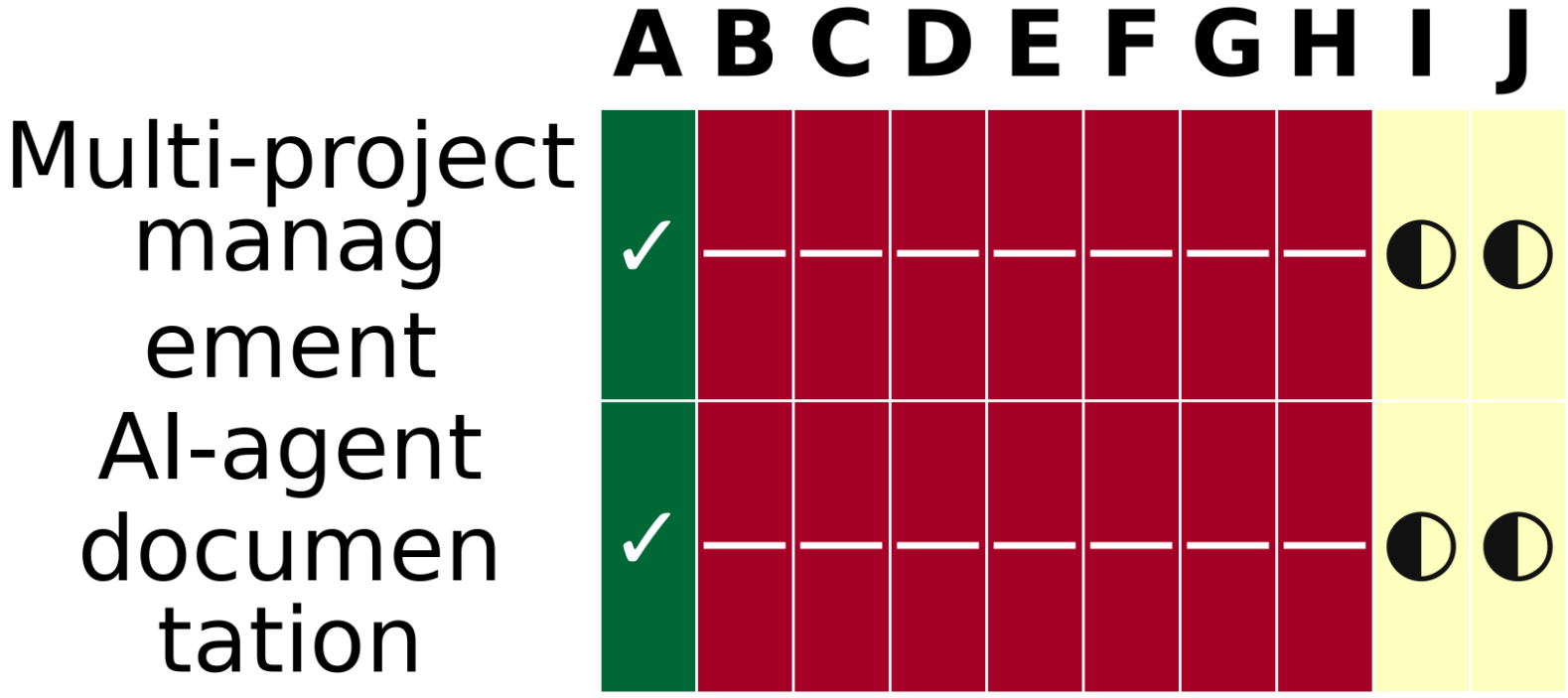



✓ full · ◐ partial · — absent

# Capabilities 4–6 · tools A–J

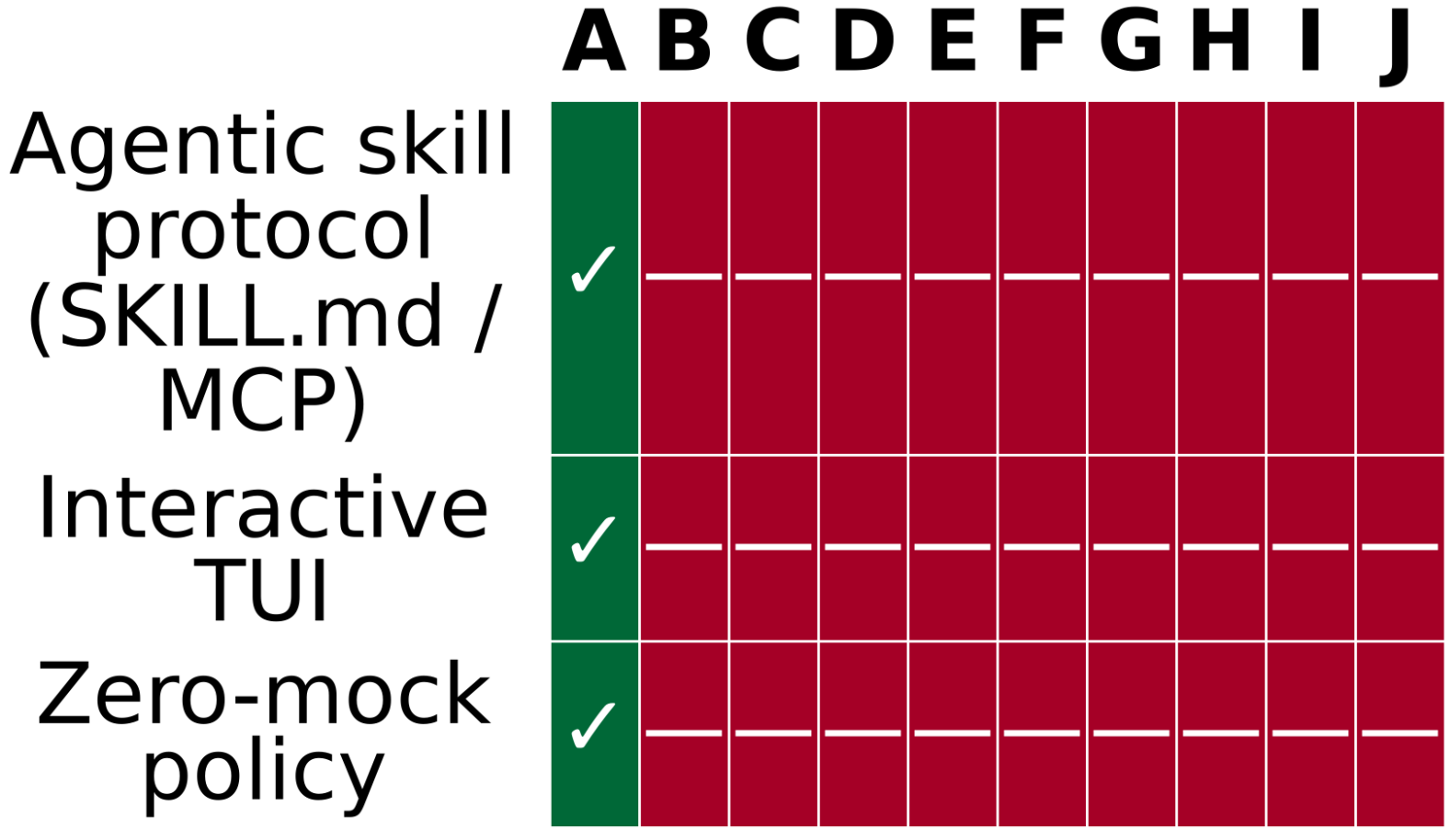


# Capabilities 7–8 · tools A–J

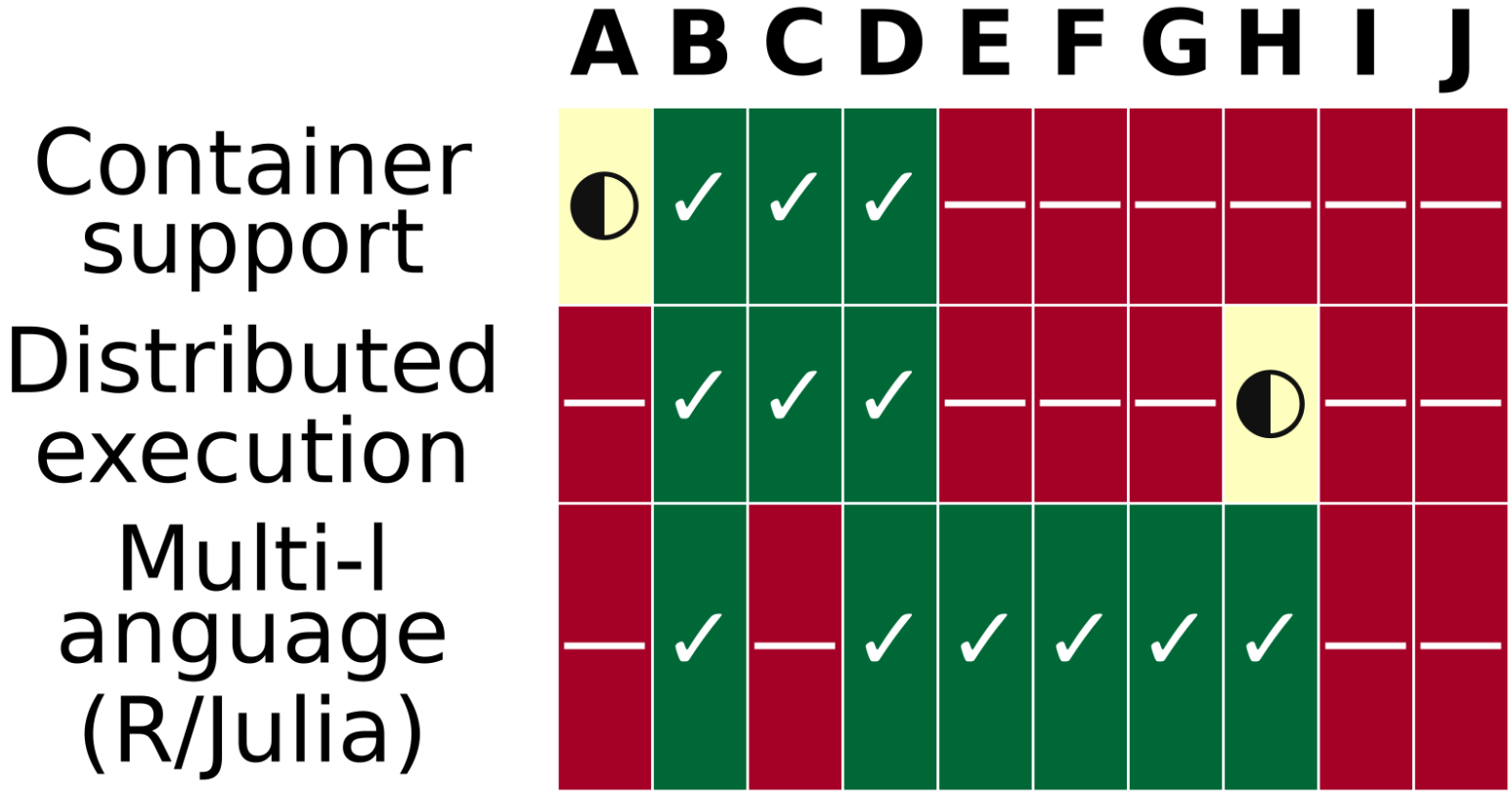



✓ full · ◐ partial · — absent

# Capabilities 9–11 · tools A–J



# Capabilities 12–14 · tools A–J




✓ full · ◐ partial · — absent

Figure 6: $14 \times 10$ heatmap annotated in appendix text—green ✓ full native capability, yellow partial / plugin-hosted, red — unavailable. Rows group *Core Pipeline*, *Quality & Security*, then *Ecosystem*.

## Comparative Feature Analysis I

[1] Nextflow lineage records exist at workflow scope, not per rendered PDF citation graph.

[2] DVC: content-addressed artifacts without native prose rendering.

[3] DVC: remote object stores (S3, GCS, Azure) without turnkey CI manuscript gates.

# Test Quality Metrics I

- **Zero mocks:** repository policy bans `unittest.mock` / patching frameworks in tests.
- **Filesystem + subprocess realism:** ephemeral directories + actual CLI binaries.
- **HTTP realism:** infra suites favour `pytest-httpserver`; literature tests hit recorded fixtures.
- `template_code_project` focuses on numerical reproducibility assertions.
- `template_autoresearch_project` exercises the AutoResearch readiness planner (`infrastructure/autoresearch/`). `template_search_project` exercises literature-search workflows.